BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 145 · Boolean Logic: AND, OR and NOT

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why should a corrected answer be retested with a different example?

Q2. What does an ethical issue concern?
A. What is fair and right
B. Only what is legal
C. Only what is popular
D. Only what is fastest

Q3. “Open source means there are never any licence conditions or costs.” Correct this. Explain why.

Q4. A computer needs maintenance and data protection. “Utilities are only apps for writing essays.” Correct this. What do utilities do?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

AND gate — Output 1 only when both inputs are 1.

OR gate — Output 1 when at least one input is 1.

NOT gate — Reverses its single binary input.

Boolean expressions — Symbols and truth tables describe logic gate relationships independently of hardware.

New learning

AND gate: Output 1 only when both inputs are 1.

OR gate: Output 1 when at least one input is 1.

NOT gate: Reverses its single binary input.

Boolean expressions: Symbols and truth tables describe logic gate relationships independently of hardware.

Worked example

For Q = A AND B, complete all four rows of the two-input truth table.

1. List A,B as 00,01,10,11.

2. AND returns 1 only if both are 1.

3. Outputs are 0,0,0,1.

4. Draw a two-input AND symbol matching the rule.

We Do · Practise with guidance

Guided 1

For Q = A OR B, complete the truth table.

Worked method: Outputs 0,1,1,1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the Boolean test that denies access when authorisation is absent.

```
authorised = false
if ... then
    print("Deny")
endif
```

write

Question 1

State AND output when A=1, B=0. [1 mark]

Question 2

Write OR outputs for inputs 00,01,10,11. [4 marks]

Question 3

Explain NOT output for input 0. [2 marks]

Exit Ticket · Answer before checking

An AND gate outputs 1 only when all its inputs are 1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. To check that the underlying method works independently on another case.

Q2. A — What is fair and right

Q3. It permits access to source code and modification or redistribution under its conditions.

Q4. Utilities maintain or protect a computer, adding tools beyond the OS. Writing an essay is an application task.

I Do / We Do pages · Guided practice

Q1. Outputs 0,1,1,1.

You Do pages · Code First

```
authorised = false
if NOT authorised then
    print("Deny")
endif
```

Expected result: Deny

You Do pages · Independent questions

Q1. 0.

Q2. 0,1,1,1.

Q3. NOT inverts the bit; output 1.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: